# A Living Review of Quantum Information in High Energy Physics

Abstract: Inspired by "A Living Review of Machine Learning for Particle Physics"[1], the goal of this document is to provide a nearly comprehensive list of citations for those developing and applying quantum information approaches to experimental, phenomenological, or theoretical analyses. Applications of quantum information science to high energy physics is a relatively new field of research. As a living document, it will be updated as often as possible with the relevant literature with the latest developments. Suggestions are most welcome.

[1]See https://github.com/iml-wg/HEPML-LivingReview.

The purpose of this note is to collect references for quantum information science as applied to particle and nuclear physics. In order to be as useful as possible, this document will continually change. Please check back[2] regularly. You can simply download the .bib file to get all of the latest references. Suggestions are most welcome.

# 1 Reviews

- Quantum Machine Learning in High Energy Physics [1]

# 2 Whitepapers

- Quantum Simulation for High Energy Physics [2]
- Quantum Computing for Data Analysis in High-Energy Physics [3]
- Snowmass White Paper: Quantum Computing Systems and Software for High-energy Physics Research [4]

# 3 Quantum Machine Learning

## 3.1 Quantum Annealing

- Solving a Higgs optimization problem with quantum annealing for machine learning [5]
- Quantum adiabatic machine learning with zooming [6]

## 3.2 Quantum Circuits

- Application of quantum machine learning using the quantum variational classifier method to high energy physics analysis at the LHC on IBM quantum computer simulator and hardware with 10 qubits [7]
- Application of quantum machine learning using the quantum kernel algorithm on high energy physics analysis at the LHC [8]

[2] See https://github.com/PamelaPajarillo/HEPQIS-LivingReview.

## References


[1] W. Guan, G. Perdue, A. Pesah, M. Schuld, K. Terashi, S. Vallecorsa et al., *Quantum Machine Learning in High Energy Physics*, 2005.08582. 1

[2] C. W. Bauer et al., *Quantum Simulation for High Energy Physics*, 2204.03381. 1

[3] A. Delgado et al., *Quantum Computing for Data Analysis in High-Energy Physics*, in *2022 Snowmass Summer Study*, 3, 2022. 2203.08805. 1

[4] T. S. Humble et al., *Snowmass White Paper: Quantum Computing Systems and Software for High-energy Physics Research*, in *2022 Snowmass Summer Study*, 3, 2022. 2203.07091. 1

[5] A. Mott, J. Job, J. R. Vlimant, D. Lidar and M. Spiropulu, *Solving a Higgs optimization problem with quantum annealing for machine learning*, *Nature* **550** (2017) 375–379. 1

[6] A. Zlokapa, A. Mott, J. Job, J.-R. Vlimant, D. Lidar and M. Spiropulu, *Quantum adiabatic machine learning with zooming*, 1908.04480. 1

[7] S. L. Wu et al., *Application of quantum machine learning using the quantum variational classifier method to high energy physics analysis at the LHC on IBM quantum computer simulator and hardware with 10 qubits*, *J. Phys. G* **48** (2021) 125003, [2012.11560]. 1

[8] S. L. Wu et al., *Application of quantum machine learning using the quantum kernel algorithm on high energy physics analysis at the LHC*, *Phys. Rev. Res.* **3** (2021) 033221, [2104.05059]. 1